Efficacy of Orion Flywheel Exercise Training in Ambulatory Subjects

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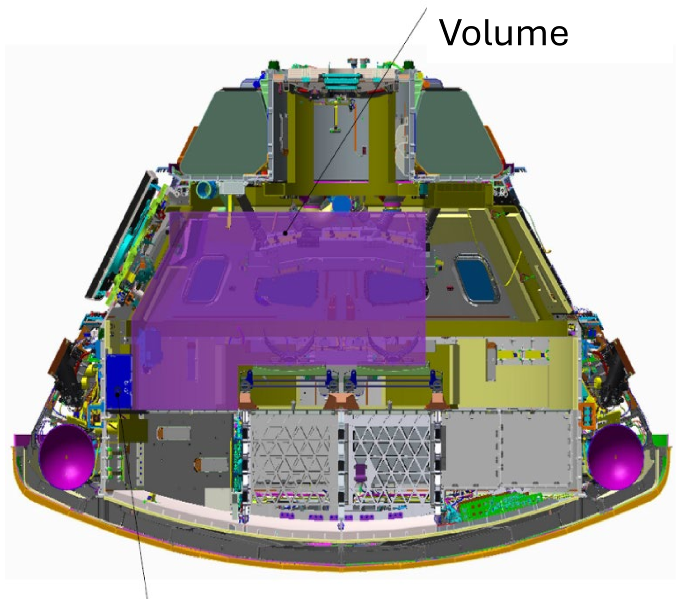
Background

The Mission

- Artemis will recommence human exploration of the Moon
- The Orion vehicle will transport crewmembers to the Moon and back
- It can support crewmembers for up to 30 days at a time
- Artemis crewmembers will be exposed to microgravity for up to 12 days before landing on the moon to perform surface EVAs

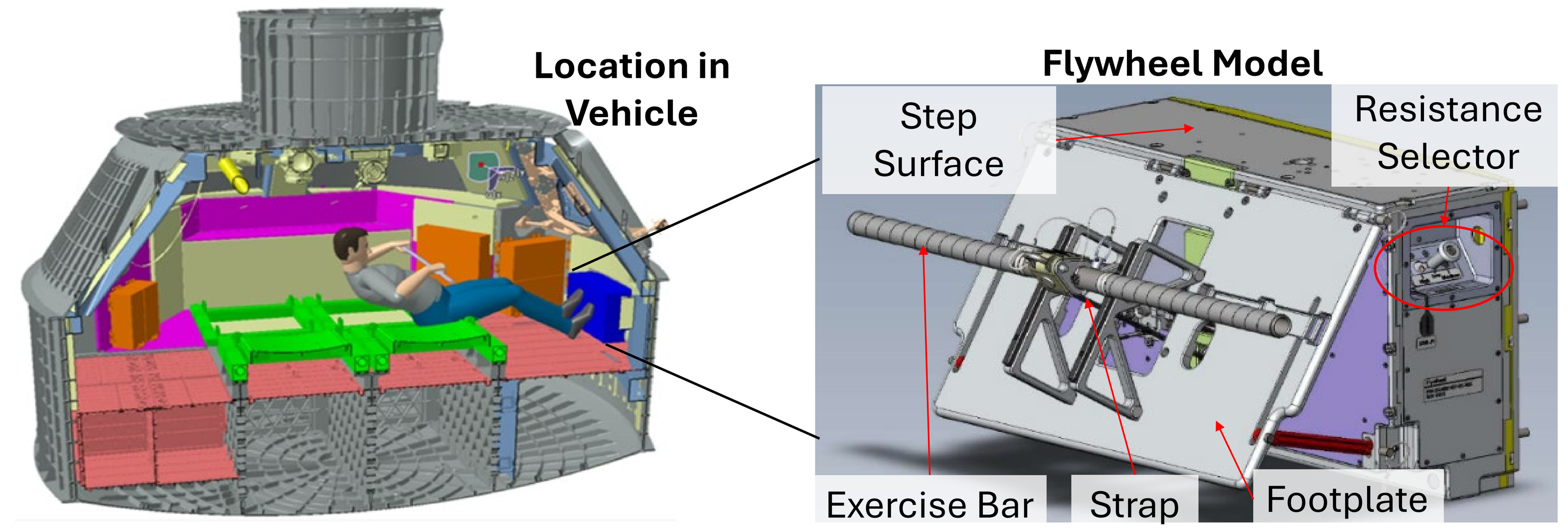
The Risk

- Even with a robust suite of exercise countermeasures such as those on the ISS, crewmembers can still lose %20+ of their preflight fitness and functional performance ability
- Artemis missions pose a greater health and performance risk than Apollo or microgravity missions
  - Longer duration than Apollo
  - Greater physical demands than Apollo or ISS
- Health and performance countermeasures to spaceflight are *significantly* limited by cabin volume and vehicle launch mass
- A compact but capable exercise device is needed in Orion to mitigate spaceflight deconditioning



The Countermeasure

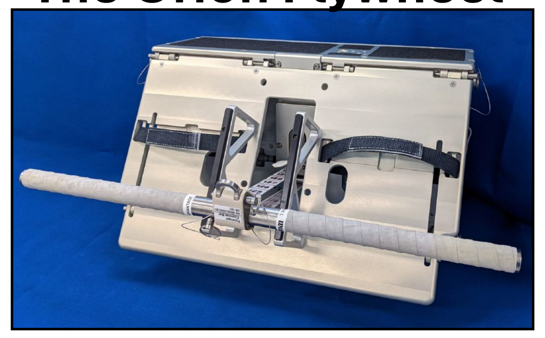
- The Orion Flywheel is a novel exercise device for spaceflight
  - Unique Load profile
  - Use of rowing ergometry for aerobic exercise
- The Flywheel will be hard-mounted to the wall of the Orion cabin
- Artemis exercise requirements specify exercise 6 days per week, alternating each day between resistance and aerobic modalities



The Orion Flywheel

- The Orion Flywheel uses "iso-inertial" loading; resistance is provided as the user accelerates and decelerates a spinning mass
- Internal gears allows the user to select from three different resistance levels
- Resistance exercise selection is limited to lower body and pulling exercises
- Combined aerobic and resistance prescriptions on a flywheel device have not been well-studied
- It is uncertain whether the Flywheel can be used at the volume expected for Artemis exercise prescriptions

The Orion Flywheel



Deadlift Exercise

Rowing Exercise

Aims of the Study

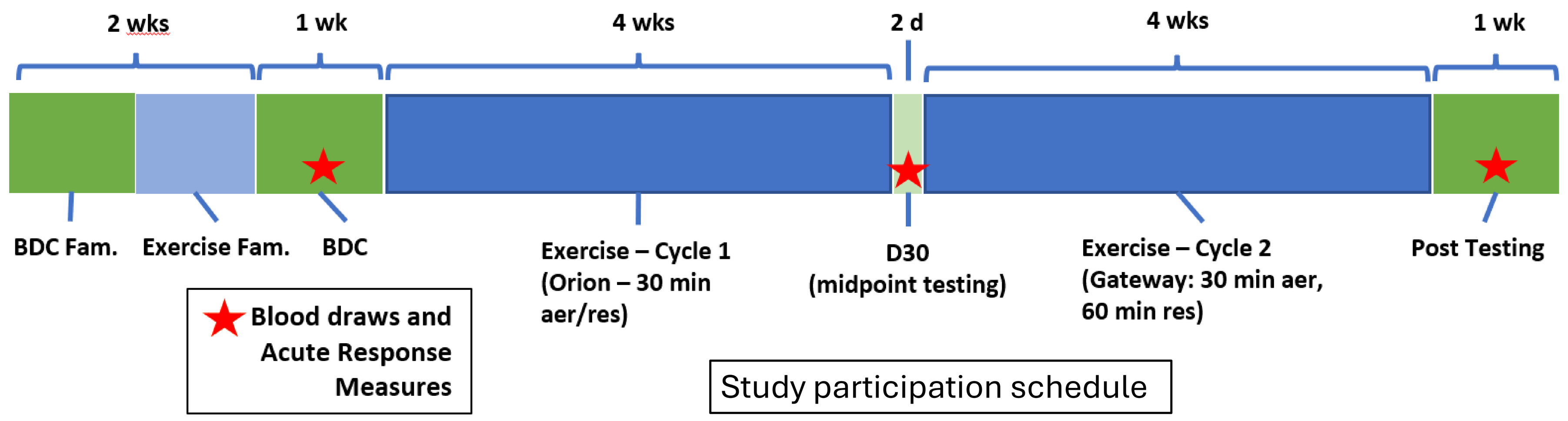
Objective

Evaluate the efficacy of the Orion Flywheel, as determined by relevant physiological health and performance outcomes, for promoting improvements in aerobic capacity, full-body muscular fitness, and functional task performance across 8 weeks of exercise

Over-arching questions

- Can the flywheel support the exercise frequency and duration required?
- How well can flywheel-only exercise preserve the various aspects of fitness and performance?

- Aim 1: Evaluate the efficacy of the Orion Flywheel for promoting improvements in aerobic capacity, full-body muscular fitness, and functional task performance across 8 weeks of exercise
- Aim 2: Evaluate subject exercise tolerance to Orion Flywheel exercise prescriptions over the course of 8 weeks within the parameters of Artemis exercise time requirements



Study participation schedule

Methods

Approach

- NASA and UHCL researchers will collaborate to implement a controlled exercise intervention study to investigate the acceptability and efficacy of the Orion Flywheel as a stand-alone exercise system
- Subjects will participate in the 13-week study, including familiarization, Pre/Mid/Post testing, and two 4-week exercise training phases.
- Subjects will exercise 6 days/week, alternating between aerobic and resistance modalities for 30-minute sessions. The second 4-week phase increases resistance session time to 40 minutes.
- A battery of fitness, performance, and exercise response measures to assess acute and chronic outcomes of Flywheel exercise

UHCL Health and Human Performance Institute

Subjects

- 16 active, healthy subjects will be recruited to participate in the 8-week exercise intervention
- Subjects will be recruited from the local community and will complete physical activity questionnaires to screen for exercise history

Measures

- Outcome measures will be assessed at baseline, at the mid-point of the exercise intervention after the first 4 weeks of training, and at the end of the 8-week exercise intervention
- Protocols included will assess full-body fitness changes using standard laboratory tests, as well as full-body functional performance using a battery of EVA-relevant tasks
- Blood and saliva will be collected to assess immune system response to Flywheel exercise as well as muscle damage and the inflammatory response to Flywheel exercise

Statistical Analysis

- Data will be analyzed using generalized linear mixed models including subject-specific random effects to address the repeated measures design
- Three timepoints (pre, mid, and post) will be included in the model as a categorical fixed effect

Status and Schedule

- The Flywheel exercise device is being built by the NASA engineering team
- Recruitment for study participation will begin in Summer 2026
- Data collection will begin in August, upon delivery and check-out of the Orion Flywheel exercise device